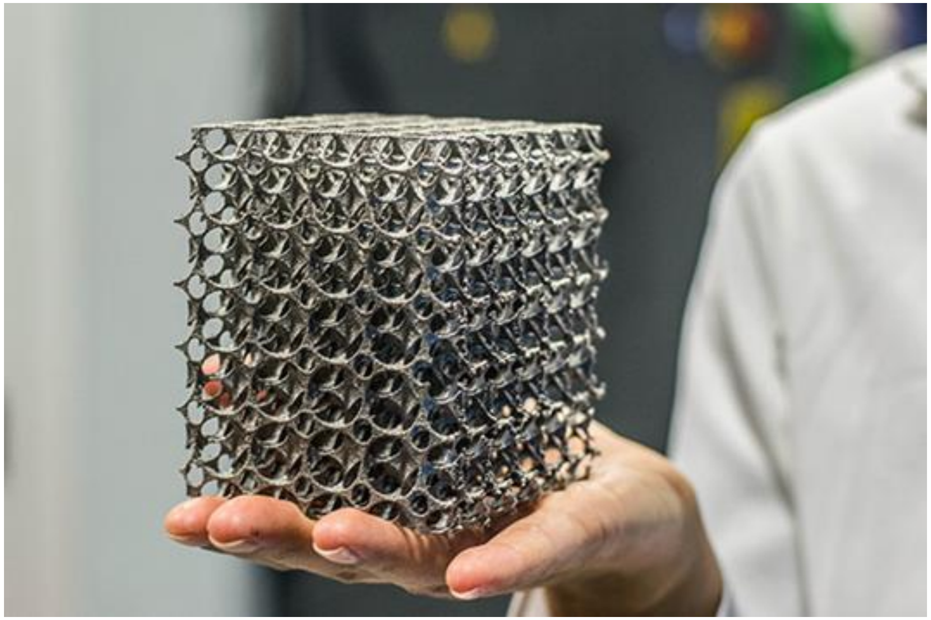
Lecture 12 – Additive Manufacturing

aka Rapid Prototyping
or 3D Printing
or Digital Manufacturing



James Busfield

The Economist

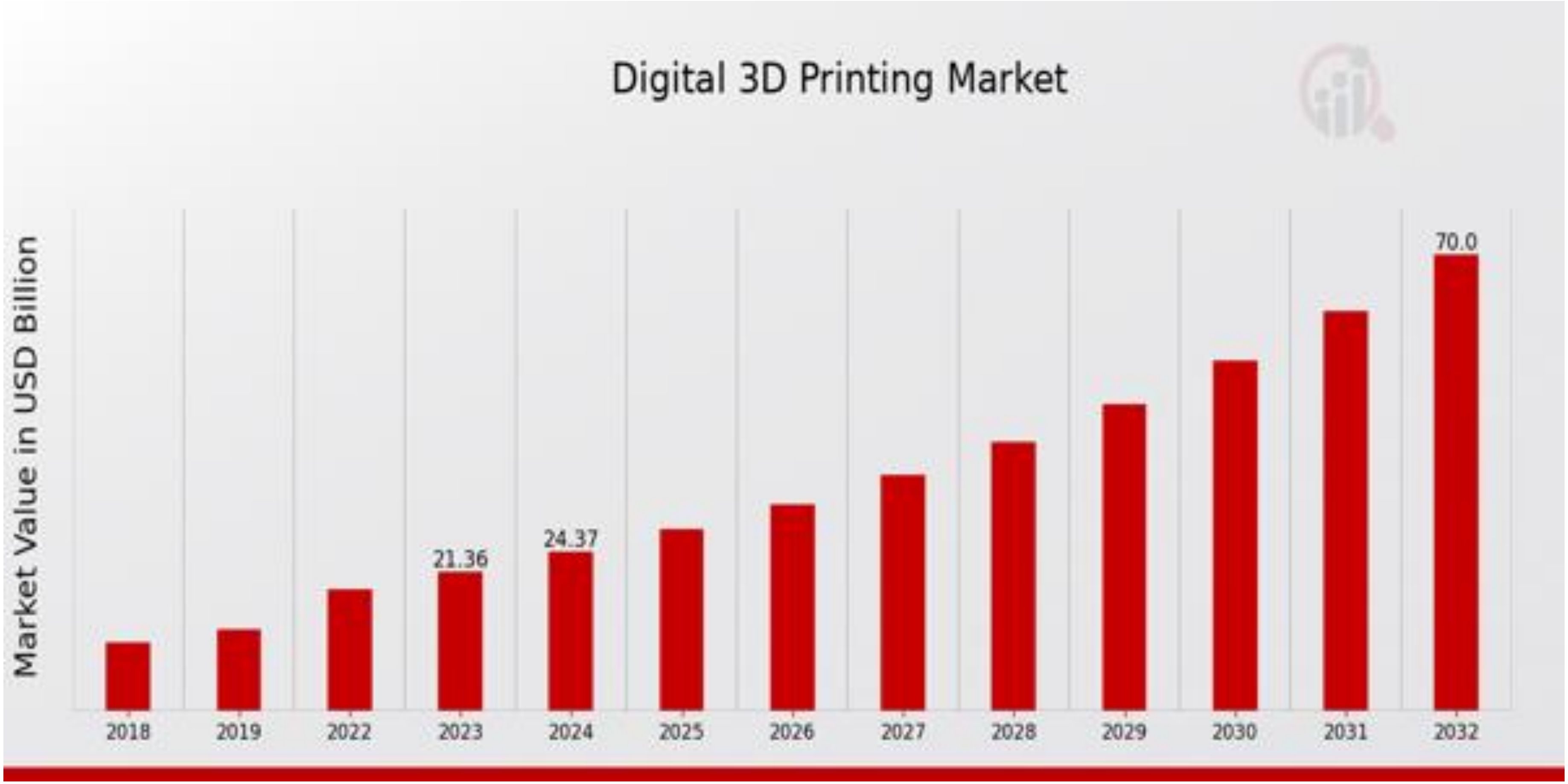
Digital manufacturing: There is a lot of hype around 3D printing. But it is fast becoming integrated with mainstream manufacturing.

3D Printing of a Boat

Queen Mary University of London

The Role for Additive Manufacturing 1

- The market for 3D printers and services was $2.2 billion worldwide in 2012 and was $21.4 in 2023. Projected to be over $70 by 2032.



https://www.marketresearchfuture.com/reports/digital-3d-printing-market-8680

The Role for Additive Manufacturing 2

- This technological shift enables manufacturers to produce highly complex and customized parts with lower material waste.
- Bioprinting technology is opening new frontiers in the medical field, making it possible to create tissues and organs that closely mimic natural biological structures.
- Moreover, the integration of artificial intelligence (AI) and machine learning in 3D printing processes is enhancing efficiency and reducing operational costs.

The role for additive manufacturing 2

- One US airline was frequently having to ground its ageing McDonnell Douglas MD-80 jets because of leaking toilets. Production of spare parts ceased long ago. Its plumbing is now 3D printed in an aerospace-grade plastic (which does not ignite or produce noxious fumes if burned).
- Millions of hearing-aid shells have already been 3D-printed from scans of patients' ear canals.
- The F-35, has around 900 parts that have been identified as suitable for additive manufacturing.
- A Chinese short-haul jet manufacturer uses giant 3D-printing machines, one of them 12 metres long, to print parts (including wing spares and fuselage frames) in titanium.
- ……….. early patents on fused-deposition modelling expired in 2009. This is what has brought the price of some printers down to below $1,000.

Rapid Prototyping

- Historically model makers employ methods like numerical control (NC) machining or hand modelling from clay, wood, or plastic foam to prepare prototypes.
- To reduce costs and speed up model-building operations, a number of fast prototyping techniques have emerged. These technologies transform CAD geometry quickly into solid objects.
- It allows an enhanced visualisation capability for a designer who can have a 3-D solid object to hold, measure, evaluate, validate and now to even use and sell.

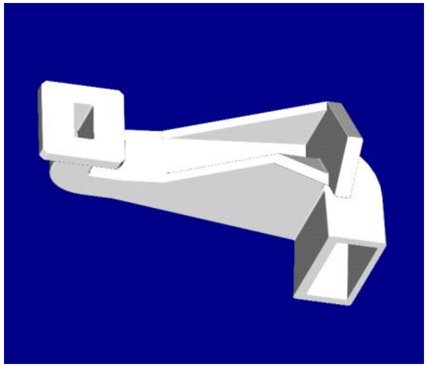
Other benefits of using prototypes include:

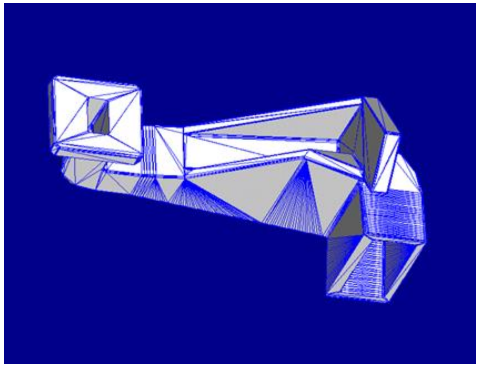
- Rapid speed of manufacture
- High level of accuracy, and
- It leads to the elimination of design flaws.

Typical Products

The Essential Process

Solid CAE model is created → **Convert to .sta file** →





The Essential Process

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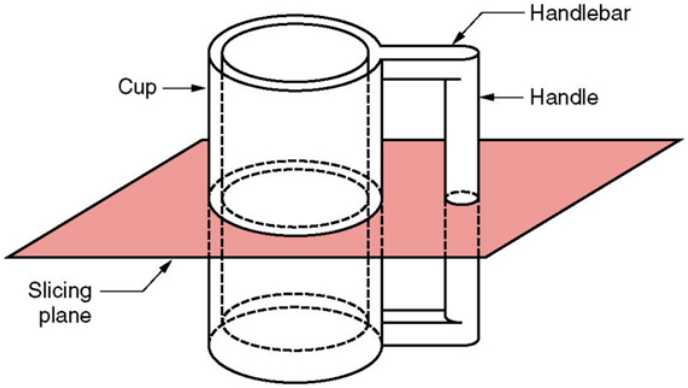
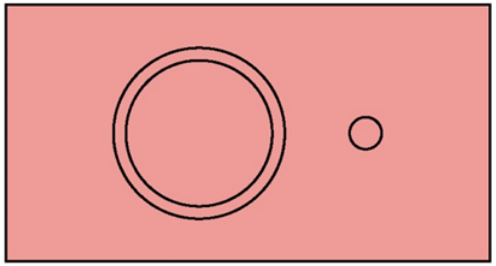
Solid CAE model is created → **Convert to .sta file** → **Create a .sli file**



(a)



(b)

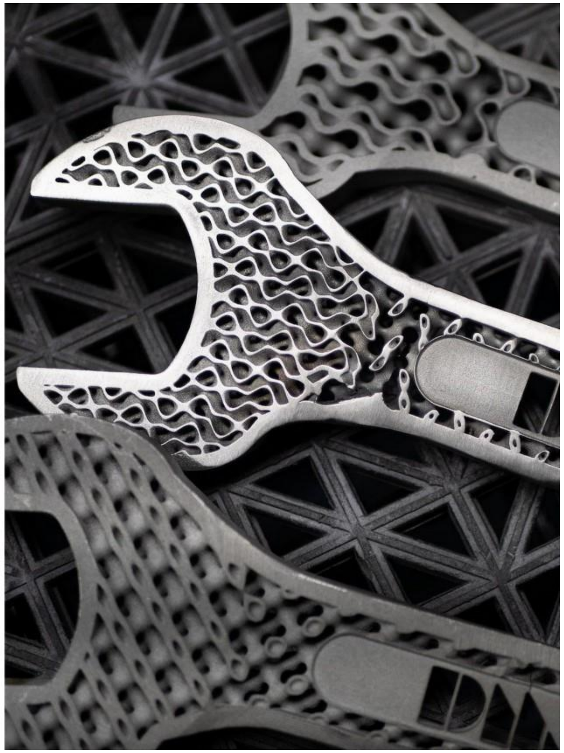
containing a series of closely spaced 2D cross-sections

The Essential Process

Solid CAE model is created

Convert to .sta file

Create a .sli file



containing a series of closely spaced 2D cross-sections

Create build file

Introduction Video

Selective Laser Sintering of Polymers

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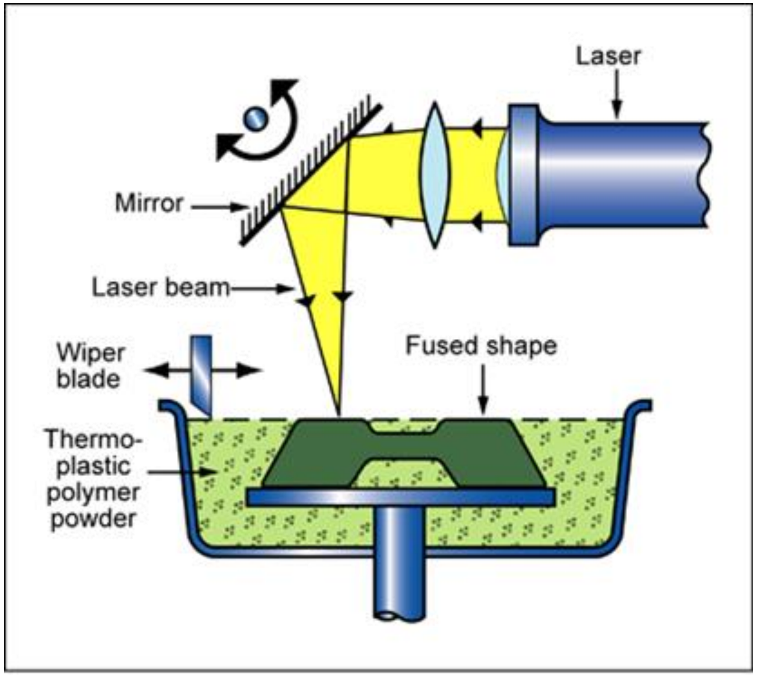
●Here powdered materials such as ABS, nylon, polycarbonate, and investment casting wax are transformed into solid objects, again a thin cross-section at a time, using a modulated laser beam.



1. A thin layer of heat-fusible powder is deposited into a workspace container. This layer is heated to just below its melting point.

2. The first cross-section of the object under fabrication is traced on the layer of powder by a heat-generating CO2 laser. which fuses the powder particles and forming a solid mass.

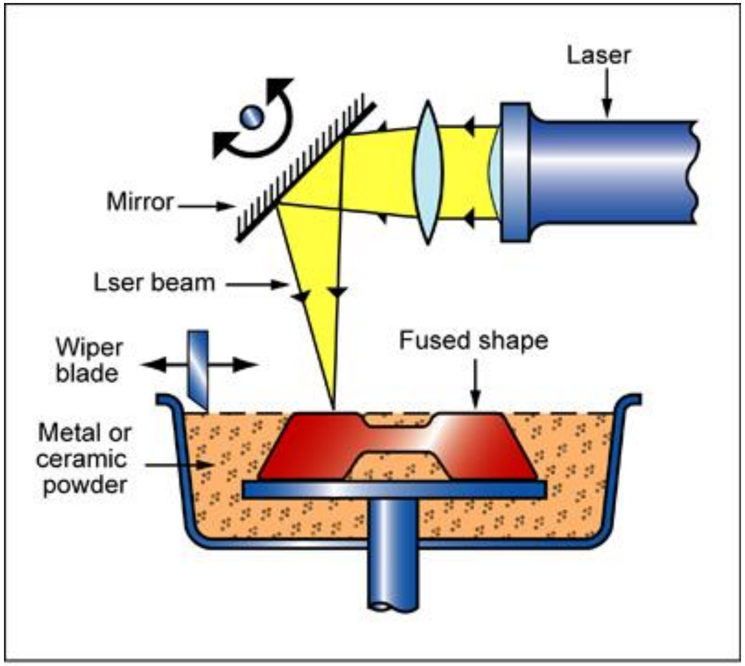
3. Another layer of powder is deposited into the workspace, on top of the previous layer and the process repeats.

●Laser sintering system is versatile, accurate, and fast, and is widely used by a range of industries for fluid and motion control parts for airframes, housings, and handles for power tool products, testable circuit breaker components, and emergency vehicle spotlight assemblies.

SLS Machine

Selective Laser Sintering of Metals

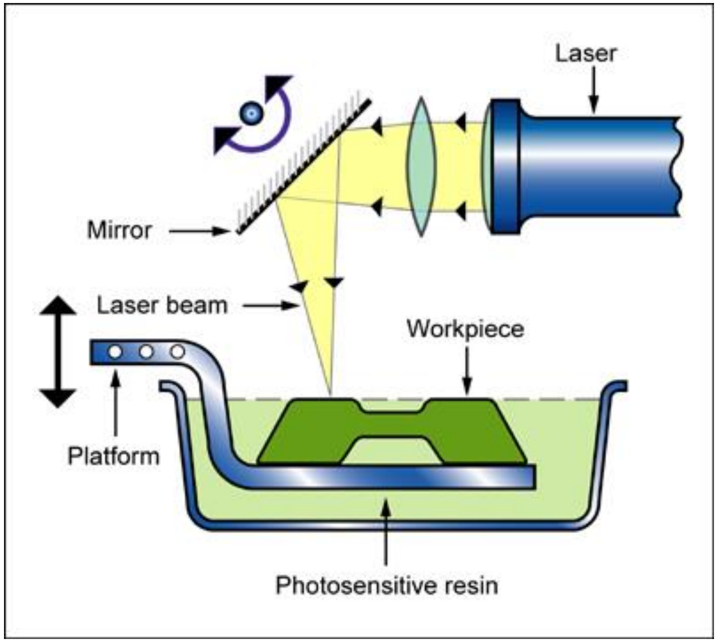
- The uppermost layer of a bed of powdered metal such as stainless steel is loosely bond through slight sintering by a scanned laser beam.
- A new layer of powder is swept across the surface and the process repeated, building the model layer by layer.
- When complete, the very porous model is infiltrated with liquid bronze, which wets and is drawn into the porosity giving a fully dense product.
- The part can have a complex external and internal shape.
- It can include cooling channels that can be used as a mould for injection moulding for example and die casting.



Metal Laser Sintering

Stereo-lithography 1

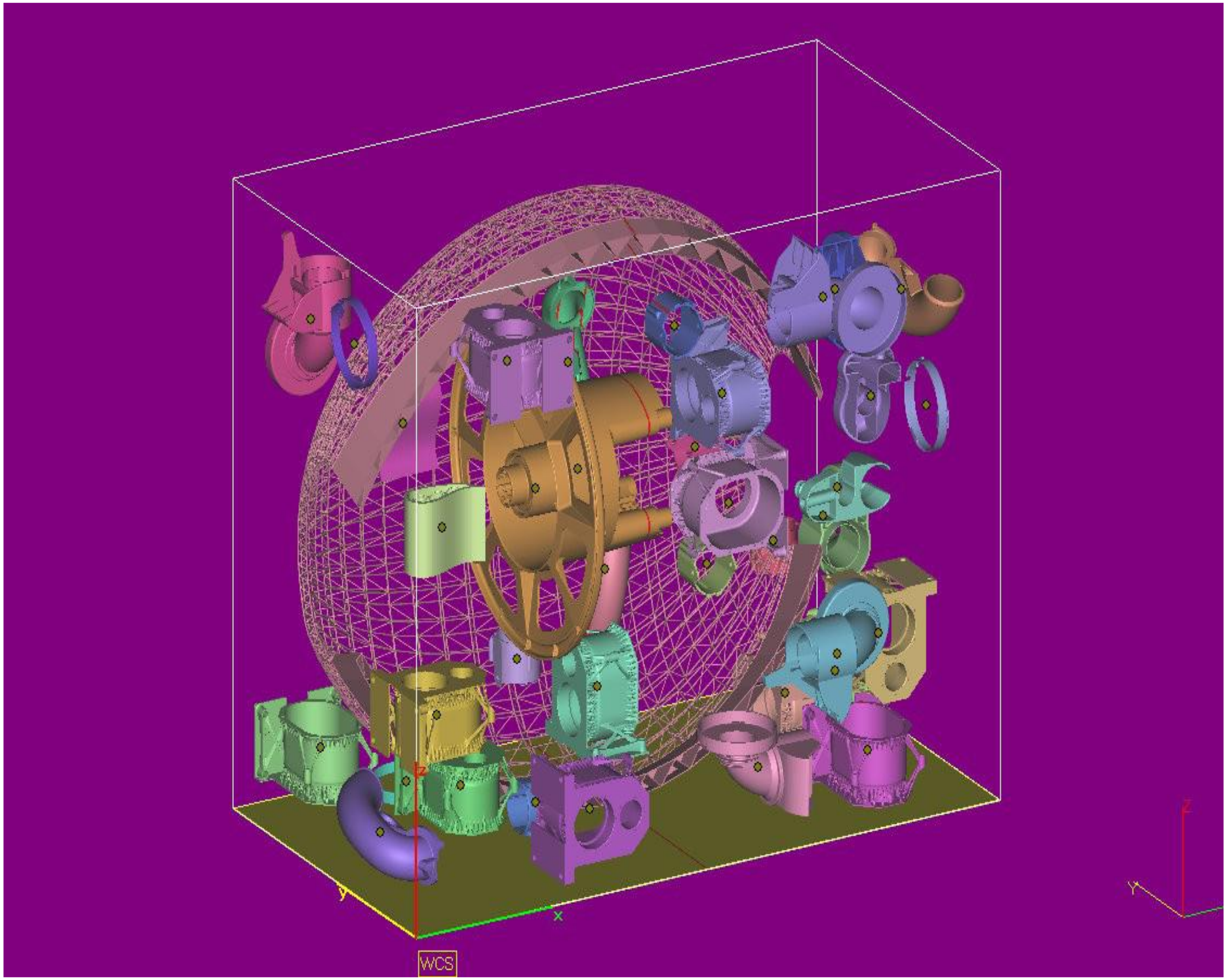
- Stereo-lithography was the first modern additive manufacturing technique, invented in 1986 by Chuck Hull of 3D Systems.
- It exploits photopolymers, mostly based on acrylates, that change from liquid to solid when exposed to ultraviolet light.



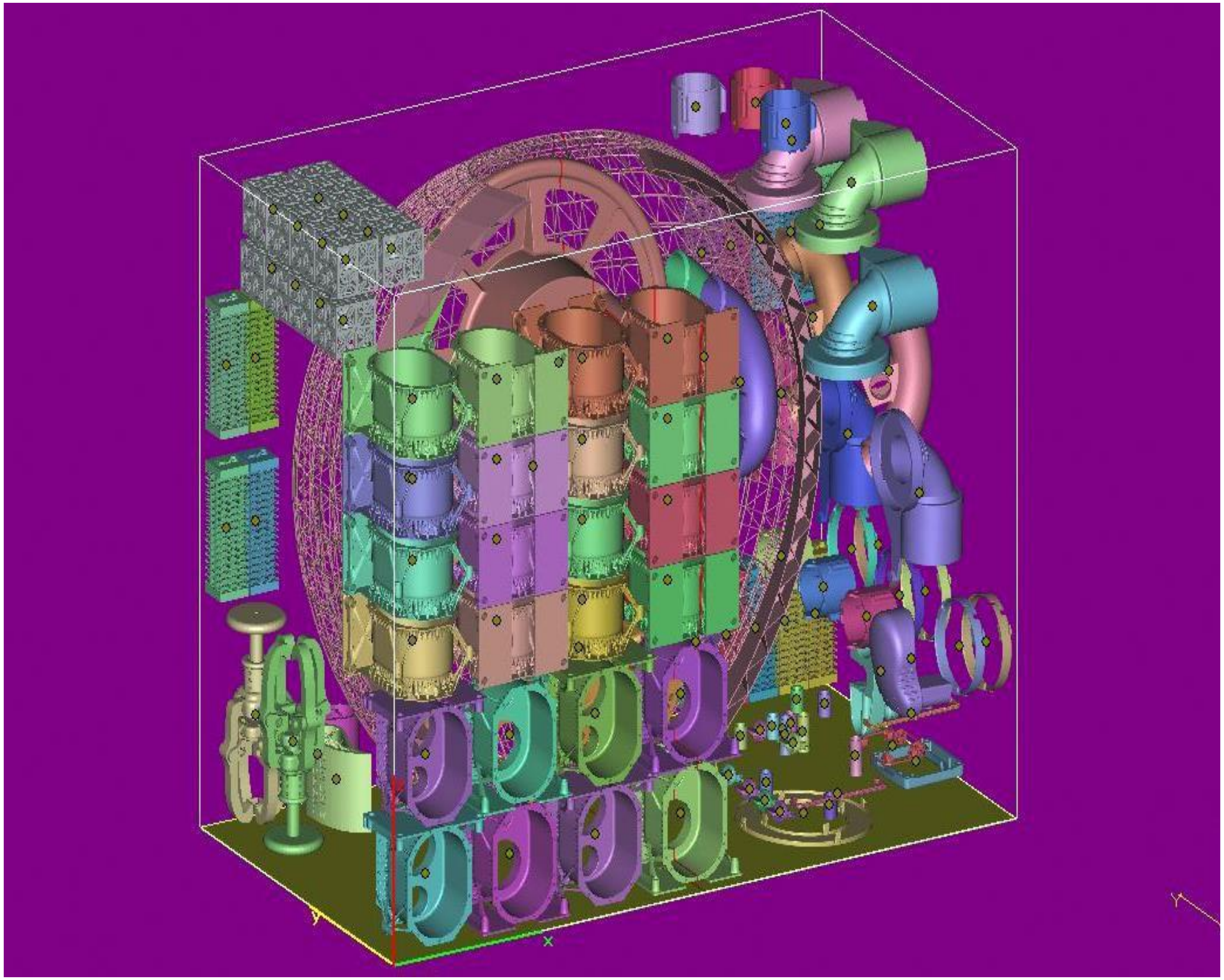
1. A laser beam guided by a mirror traces a single layer onto the surface of a vat of liquid polymer to cure the polymer.
2. The cured material is then lowered a small distance (typically about 0.1mm) below the remaining liquid, re-coating it with more resin and the process is repeated.
3. Post curing is required to full solidify the part.

- SLA product applications include medical bone implants, electrical discharge machining electrodes, nozzles for metal spraying, cellular phone cases, and automotive wheel rims.

Partially Packed Chamber



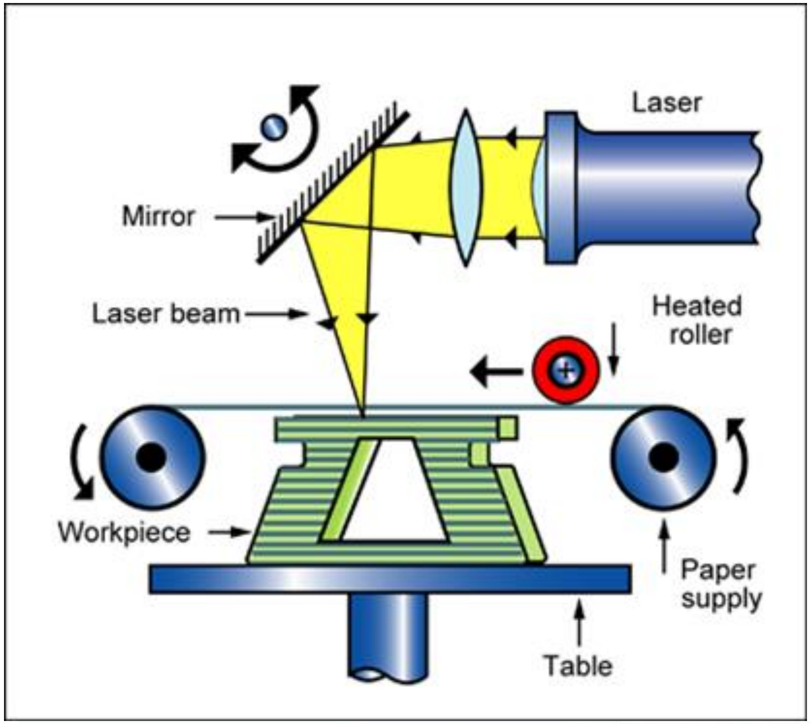
Fully Packed Chamber

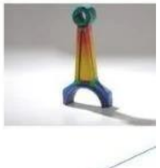


Removing Parts from build chamber

Laminated Object Manufacturing (LOM)

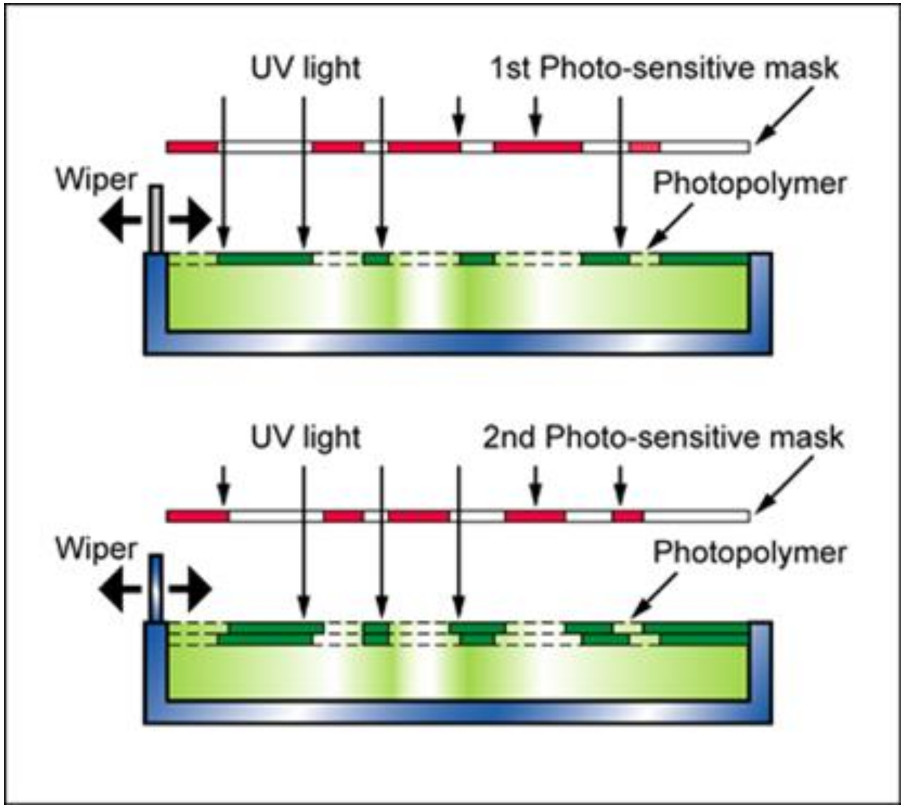
- LOM is used to produce solid objects by successive deposition, bonding, and laser cutting of layers of sheet materials such as papers, polymer or even metal, or composites.
- A coated feed materials is fed from a roll, positioned over the model, and cut to shape by a guided laser beam. The cut profile is pressed onto the model by a heated roller, bonding it to the layers beneath.
- The sequence is repeated.
- Part can be used as moulding and casting patterns, and limited test or visual models for the automotive, aerospace, medical, computer, and electronics industries.

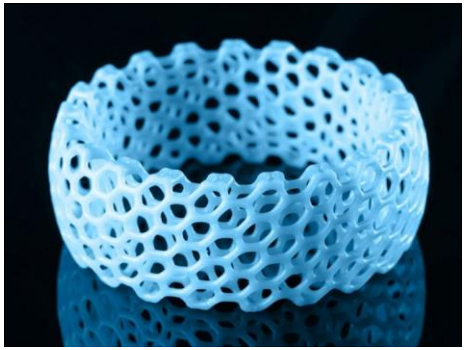




Solid Ground Curing (SGC)

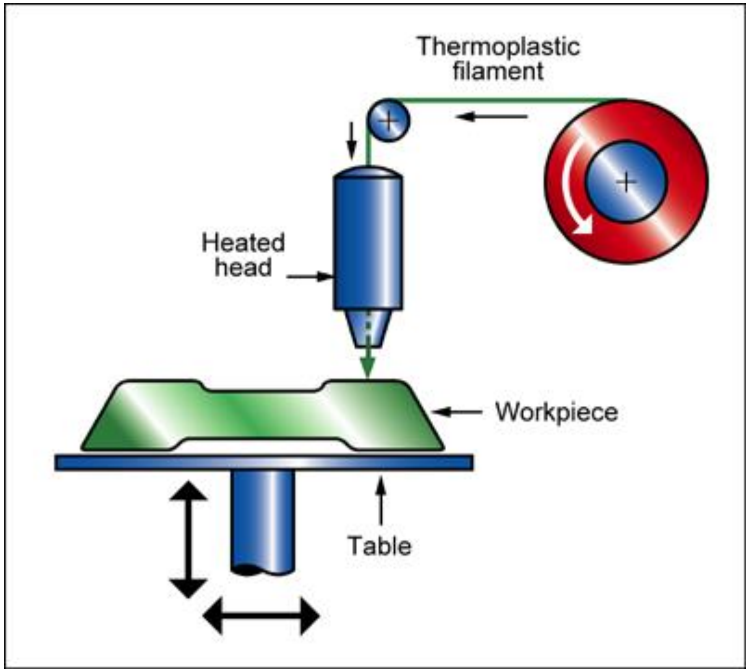
- SGC creates a negative image of the layer of the part to be electrostatically on a glass plate (as in a photocopier).
- This mask is positioned over a bath of photo-sensitive resin and the clear areas cured by exposure to UV light.
- The resin is wiped clear, leaving hardened areas intact. Wax is applied to the top surface to provide support to the cured resin, and solidified by chilling. The model is then lowered by one layer thickness and the next layer of resin is applied for curing.
- The top surface is then milled back to the required thickness and the next layer of resin is applied for curing.

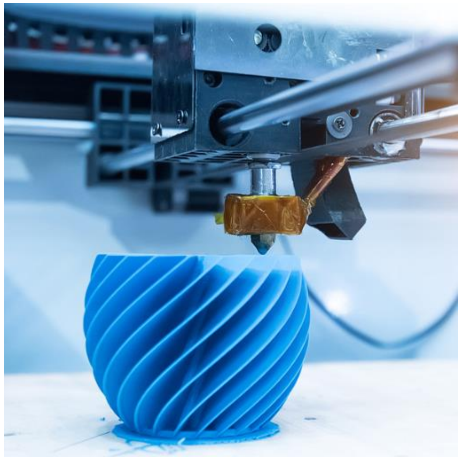




Fused Deposition Modelling (FDM)

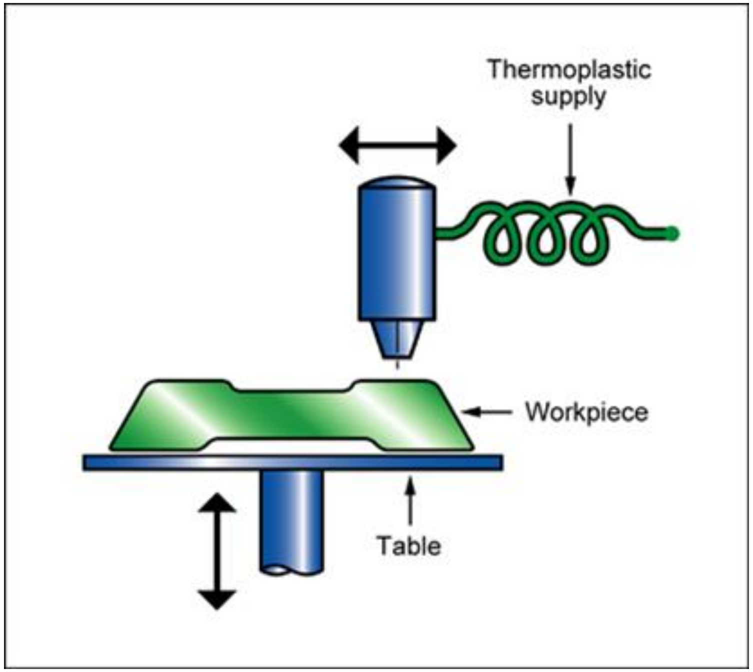
- A fine stream of thermoplastic (wax or metal could also be used) is deposited by a two-axis heated extrusion head.
- This material is extruded and then deposited into layers, typically 0.1 mm, one layer at a time starting at the base. This builds the model vertically on a fixtureless base. Successive layers adhere together through thermal fusion.
- The FDM process requires no post curing enabling multiple versions of a part to be created within a short time frame.
- Materials typically available for use in the process include ABS and Nylon.





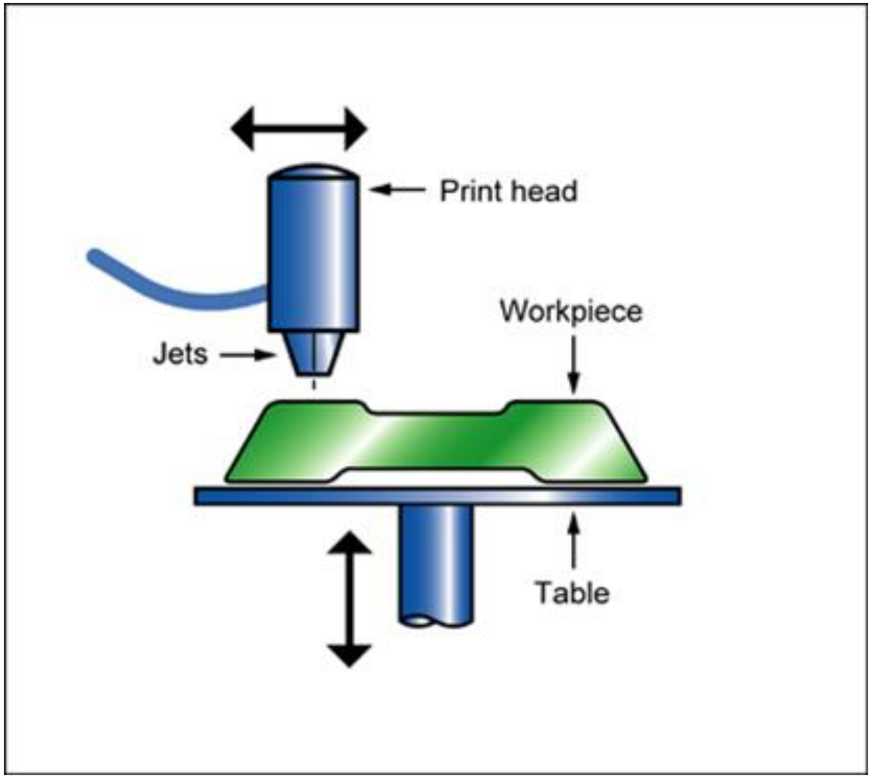
Ballistic Particle Manufacture (BPM)

- BPM is a rapid prototyping technique in which microscopic particles of molten thermoplastic are shot by a piezoelectric jetting system and freeze when they hit the object being created.
- The only finish available is in white. No material is wasted in this process. A wide range of materials can be used.



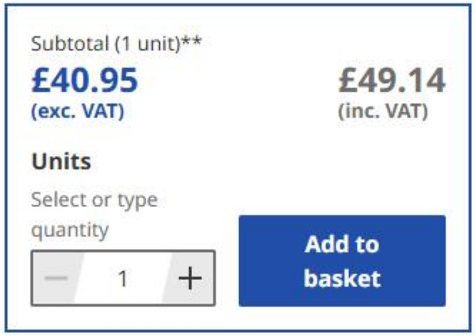
3D Printing

- (3D printing) draws on the technology of ink-jet printers to build up successive layers of a prototype or model.
- Instead of ink, the print-head deposits thermoplastic polymer that quickly sets.
- A print head of 96 jets orientated in a linear array gives a print density comparable to a 300dpi printer.
- The attraction of 3-dimensional printing is that it uses multiple jets, greatly increasing the speed at which the model can be built.



Current problems for additive manufacturing

- It can be slow - taking several hours to print, say, a body panel for a car. Too too slow for large production runs?
- It is perfect though for one-off items which would take weeks to make in a machine-shop.
- Material costs are also high. ABS, is the most common 3D-printing material. A mass manufacturer using plastic injection moulding might buy ABS in bulk for about £1.77 /kg (2025 Alibaba Prices). A bespoke powder or filament for 3D printing costs £67 /kg. (2025 RS prices)



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